

Daily Geometry Drill #6 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

Section A Rapid Recognition

1. Answer: 40

Method: The angle at the centre is twice the angle at the circumference on the same arc: $2 \times 20^\circ = 40^\circ$.

Investigate further: Halving instead of doubling; the centre angle is the bigger one.

2. Answer: 35

Method: $\angle PQR$ and $\angle PSR$ stand on the same chord PR , so they are equal.

Investigate further: Watching the wrong chord; both angles must look at PR .

3. Answer: 110

Method: Opposite angles of a cyclic quadrilateral sum to 180° : $\angle C = 180^\circ - 70^\circ = 110^\circ$.

Investigate further: Using $\angle B$ (adjacent, not opposite) instead of $\angle A$.

4. Answer: 40

Method: The tangent–chord angle equals the angle in the alternate segment subtended by that chord, so it is 40° .

Investigate further: Quoting the angle's supplement 140° ; the equal angle is on the far side of the chord.

5. Answer: 4

Method: Intersecting chords: $PA \cdot PB = PC \cdot PD$, so $2 \cdot 6 = 3 \cdot PD$, $PD = 4$.

Investigate further: Equating $PA = PD$; the products, not the pairs, are equal.

6. Answer: 9

Method: Tangent–secant power (Day 5): $PT^2 = PA \cdot PB$, so $9 = 1 \cdot PB$, $PB = 9$.

Investigate further: Forgetting that PAB is P – A – B : PB is the full secant, not the chord $AB = 8$.

7. Answer: 90

Method: Thales: a diameter subtends a right angle at the circumference.

Investigate further: Answering 180° (the straight diameter line) rather than the right angle it subtends.

8. **Answer:** 12

Method: Secant–secant power: $PA \cdot PB = PC \cdot PD$, so $4 \cdot 6 = 2 \cdot PD$, $PD = 12$.

Investigate further: Exchanging a near point for a far point; $PB = 6$ is the *full* P – A – B segment.

9. **Answer:** 16

Method: $PT^2 = PA \cdot PB$: $64 = 4 \cdot PB$, $PB = 16$.

Investigate further: Quoting the chord $AB = PB - PA = 12$ instead of the full PB .

10. **Answer:** 30

Method: $\angle BAC$ and $\angle BDC$ stand on the same chord BC , so both are 30° .

Investigate further: Adding the diagonal's split angle; only the chord BC matters.

Section B Manipulation Drills

1. **Answer:** B) 65

Method: Half the central angle: $\frac{1}{2} \cdot 130^\circ = 65^\circ$.

Investigate further: Answering the centre angle itself (130° , option D).

2. **Answer:** A) 80

Method: Opposite angles are supplementary: $180^\circ - 100^\circ = 80^\circ$.

Investigate further: Keeping the angle as-is because the quadrilateral looks symmetric.

3. **Answer:** A) 1

Method: $3 \cdot 5 = 15 \cdot PD$, so $PD = 1$.

Investigate further: Multiplying 3 by 5 and dividing by the nearer point 5, getting a mirror of the given PC .

4. **Answer:** $\frac{25}{2}$

Method: $PT^2 = PA \cdot PB$: $25 = 2 \cdot PB$, $PB = \frac{25}{2}$.

Investigate further: Answering 5 (the tangent half) or dividing 25 by PB instead of PA .

5. **Answer:** B) 55

Method: Alternate segment theorem: the tangent–chord angle equals the angle in the alternate segment.

Investigate further: Taking $90^\circ - 55^\circ = 35^\circ$ (a tangent–radius reflex).

6. **Answer:** A) 6

Method: $PT = \sqrt{PA \cdot PB} = \sqrt{2 \cdot 18} = \sqrt{36} = 6$.

Investigate further: Adding segments into $\frac{PA+PB}{2} = 10$ rather than taking the geometric mean.

7. **Answer:** C) 36

Method: Opposite angles sum to 180° : $2x + 3x = 5x = 180^\circ$, $x = 36$.

Investigate further: Setting $2x = 3x$ or forgetting the opposite pairing.

8. **Answer:** 35

Method: $\angle ACB = 90^\circ$ (diameter), so $\angle BAC = 180^\circ - 90^\circ - 55^\circ = 35^\circ$.

Investigate further: Answering 55° as if the triangle were isosceles.

9. **Answer:** A) 4

Method: Pitot: $AB + CD = BC + DA$, so $DA = 5 + 6 - 7 = 4$.

Investigate further: Pairing adjacent sides instead of opposite ones.

10. **Answer:** 8

Method: $PA \cdot PB = PC \cdot PD$: $6 \cdot 4 = 3 \cdot PD$, $PD = 8$.

Investigate further: Treating PB as a single piece rather than P – A – B ordering.

Section C Substitution & Structure

1. **Answer:** B) 50

Method: The tangent–chord angle equals the angle in the alternate segment, which is half $\angle AOB$: $\frac{1}{2} \cdot 100^\circ = 50^\circ$.

Investigate further: Doubling to 80° or subtracting from 90° .

2. **Answer:** A) 9

Method: $PA \cdot PB = PC \cdot PD$ gives $3 \cdot 12 = 4 \cdot PD$, $PD = 9$.

Investigate further: Multiplying $3 \cdot 12$ then dividing by PC but quoting $PB = 12$ (option C).

3. **Answer:** A) 30

Method: $\angle BDC$ and $\angle BAC$ stand on the same chord BC , so they are equal (30°).

Investigate further: Adding the two split angles to get 70° , or using the 40° side.

4. **Answer:** B) 6

Method: Ptolemy: $d_1 d_2 = AB \cdot CD + BC \cdot AD = 15 + 15 = 30$, and $d_1 + d_2 = 11$ gives $(5, 6)$; larger is 6.

Investigate further: Quoting the product 30 (option C) or the sum 11 (option D).

5. **Answer:** B) 30

Method: $s = 11$; Brahmagupta: $K = \sqrt{(11-5)(11-5)(11-6)(11-6)} = \sqrt{6 \cdot 6 \cdot 5 \cdot 5} = 30$.

Investigate further: Using the perimeter 22 for s , or Heron's triangle-only formula.

6. **Answer:** B) 90

Method: Thales again: any angle standing on a diameter is right.

Investigate further: Answering the supplementary angle at the centre, which here is absent.

7. **Answer:** B) 9

Method: $PT^2 = PA \cdot PB$: $36 = 3 \cdot PB$, $PB = 12$, so $AB = PB - PA = 9$.

Investigate further: Quoting $PB = 12$ as the chord, or forgetting to subtract PA .

8. **Answer:** B) 72

Method: $\angle A + \angle C = 180^\circ$: $2x + 3x = 180^\circ$, $x = 36^\circ$, $\angle A = 72^\circ$.

Investigate further: Answering $x = 36^\circ$ (option style) instead of $\angle A = 2x = 72^\circ$.

Section D Challenge Ramp

1. **Answer:** B) 5

Method: $PC \cdot PD = 30$ with $PD = PC + 1$: consecutive factors of 30 are 5 and 6, so $PC = 5$.

Investigate further: Picking 6 (option C) by mixing up which factor is PC .

2. **Answer:** C) 6

Method: Ptolemy: $d_1 d_2 = 18 + 24 = 42$; with $d_1 = 7$, $d_2 = 6$.

Investigate further: Quoting 42 (option D) as a diagonal, or dividing by 7 and misreading.

3. **Answer:** A) $2\sqrt{6}$

Method: $s = 5$; $K = \sqrt{(5-2)(5-3)(5-4)(5-1)} = \sqrt{3 \cdot 2 \cdot 1 \cdot 4} = \sqrt{24} = 2\sqrt{6}$.

Investigate further: Forgetting to halve the perimeter, or cancelling the $(s-a)$ terms.

4. **Answer:** B) 110

Method: $\angle BAD = 30^\circ + 40^\circ = 70^\circ$, and opposite angles sum to 180° : $\angle BCD = 110^\circ$.

Investigate further: Answering the split $\angle BAD = 70^\circ$ (option A) instead of its supplement.

5. **Answer:** A) 36

Method: Sides 6, 5, 12, 5 give $s = 14$; $K = \sqrt{(14-6)(14-5)(14-12)(14-5)} = \sqrt{8 \cdot 9 \cdot 2 \cdot 9} = 36$.
Check: trapezoid height 4 gives $\frac{6+12}{2} \cdot 4 = 36$.

Investigate further: Using Heron's triangle formula, or halving the legs theorem-wise (leftover Pitot habit).

Top 5 Patterns

- **Find the chord, then compare angles on it.** Two angles subtending the same chord are equal; a central angle is twice either of them. *(see A1, A2, A10, B1, C1, C3, D4)*
- **Product power:** $PA \cdot PB = PC \cdot PD$ and $PT^2 = PA \cdot PB$. Chords crossing, or a tangent with a secant, are always a product equation. *(see A5, A6, A8, A9, B3, B4, B6, B10, C2, C7, D1)*
- **Cyclic opposites are supplementary.** $\angle A + \angle C = 180^\circ$ is the reflex for 'cyclic' claims and the lever for ratio angles. *(see A3, B2, B7, C3, C8, D4)*
- **Ptolemy adds products down opposite sides.** $AC \cdot BD = AB \cdot CD + BC \cdot AD$. *(see C4, D2, D5)*
- **Brahmagupta is Heron with brothers.** The cyclic-area formula starts from the same semiperimeter s . *(see C5, D3, D5)*

Common Traps

- **Doubling the circumference angle instead of the centre one.** The centre is twice the circumference, never the other way round. *(see A1, B1, C1)*
- **Taking the tangent–radius 90° angle as the answer.** The alternate segment angle equals the tangent–chord angle, not its complement. *(see A4, B5, C1)*
- **Quoting the full secant PB when the chord AB is asked.** Subtract the near segment: $AB = PB - PA$. *(see A6, A9, C7)*
- **Applying Heron to cyclic quadrilaterals.** Heron needs a triangle; Brahmagupta's $(s - a)(s - b)(s - c)(s - d)$ owns the four-sided case. *(see C5, D3, D5)*
- **misapplying Pitot's pairing.** Pitot pairs *opposite* sides ($AB + CD = BC + DA$); a trap variant pairs adjacent ones. *(see B9, D5)*

“A circle trades in equalities; find the chord that both angles stare at, write down the product they share, and the number appears.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths